This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Review activities: MI, IS, WC, MU, AT, 82, NSA
User activities: SH, MC, GL
CODE C

DD FORM 1 SEP 67 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

P.A. ME
Other Cust
AIR FORCE-82

PROCUREMENT SPECIFICATION FF-S-107

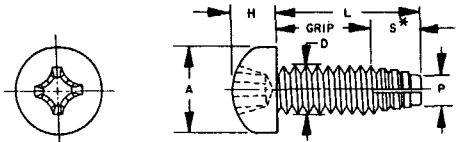
SUPERSEDES: MS24650 AND AN504

TITLE
SCREW, TAPPING-THREAD CUTTING, TYPES D,F,G OR T, PAN HEAD, CROSS RECESSED, CORROSION RESISTING STEEL

ENTIRE STANDARD REVISED

MILITARY STANDARD
MS24630

SHEET 1 OF 2



TYPE D

OPTIONAL THREAD TYPE



TYPE F



TYPE G



TYPE T

NOMINAL SIZE		2(.086)		4(.112)		6(.138)		8(.164)		10(.190)		1/4		10(.190)		1/4	
THREADS PER INCH		56		40		32		32		24		20		32		28	
MAJOR DIAMETER D	MAX	.0860		.1120		.1380		.1640		.1900		.2500		.1900		.2500	
	MIN	.0820		.1072		.1326		.1586		.1834		.2428		.1846		.2438	
HEAD DIAMETER A	MAX	.167		.219		.270		.322		.373		.492		.373		.492	
	MIN	.155		.205		.256		.306		.357		.473		.357		.473	
HEAD HEIGHT H	MAX	.062		.080		.097		.115		.133		.175		.133		.175	
	MIN	.053		.070		.087		.105		.122		.162		.122		.162	
POINT DIAMETER P	MAX	.067		.086		.106		.132		.147		.198		.158		.213	
	MIN	.061		.077		.095		.121		.133		.181		.147		.201	
LENGTH L	TOLERANCE	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN	DASH NO.	GRIP MIN
1/8	+0 -1/32	1	.031														
3/16		2	.076	9	.069	20	.047										
1/4		3	.138	10	.106	21	.109	33	.110								
5/16		4	.201	11	.169	22	.141	34	.141	44	.135			64	.135		
3/8		5	.263	12	.231	23	.203	35	.203	45	.156	56	.169	65	.156		
1/2		6	.388	13	.356	24	.328	36	.328	46	.281	57	.244	66	.281	73	.244
5/8		7	.513	14	.481	25	.453	37	.453	47	.406	58	.369	67	.406	74	.369
3/4				15	.606	26	.578	38	.578	48	.531	59	.494	68	.531	75	.494
7/8	+0 -3/64			16	.716	27	.688	39	.688	49	.641	60	.603	69	.641	76	.603
1				17	.841	28	.813	40	.813	50	.766	61	.728	70	.766	77	.728
1-1/4						29	1.063	41	1.063	51	1.016	62	.978	71	1.016	78	.978
1-1/2						30	1.313			52	1.266	63	1.228	72	1.266	79	1.228

MATERIAL: STEEL, CORROSION RESISTING, IN ACCORDANCE WITH FED.STD. NO. 66, STEEL NOS. 410 OR 420 (HARDENED).
FINISH: PASSIVATED.
THREADS: THREAD TYPES TO BE IN ACCORDANCE WITH APPLICABLE REQUIREMENTS OF PROCUREMENT SPECIFICATION.
RECESS: THE RECESS SHALL BE IN ACCORDANCE WITH MS9006.
HOLE SIZES: FOR APPROXIMATE HOLE SIZE SEE MS24634. FOR 1/4-28 SEE ASA B18.6.4.
DIMENSION: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS PLUS THE DASH NUMBER, EXAMPLE: MS24630-1.
*S(TAPER LENGTH) = 2-1/2 TO 3-1/2 TIMES THE PITCH DISTANCE FOR SCREWS HAVING A THREAD LENGTH OF 8 PITCHES OR LESS. TAPER LENGHT FOR LONGER LENGTHS WILL BE 3-1/2 TO 4-1/2 TIMES THE PITCH DISTANCE.
NOTES: (1) TYPE DESIGNATIONS ARE ESTABLISHED BY THE PROCUREMENT SPECIFICATION.
(2) FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
(3) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BID.
(4) FOR AERONAUTICAL APPLICATIONS, THESE SCREWS ARE SUBJECT TO MS33749 LIMITATION AND INSTALLATION REQUIREMENTS.

FED. SUP CLASS 5305

APPROVED 5 OCTOBER 1959 REVISED (B) 25 FEBRUARY 1963 (C) 28 AUGUST 1967

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THE SCREWS COVERED BY THE FOLLOWING DASH NUMBERS LISTED IN MS24650 AND AN504 ARE CANCELLED AFTER USE THE DASH NUMBERS GIVEN IN SHEET I. THE CANCELLED SCREWS CANNOT ALWAYS REPLACE THE SUPERSEDING SCREWS AND SHOULD BE USED UNTIL EXISTING STOCKS ARE DEPLETED. USE ONLY THE SUPERSEDING SCREWS FOR DESIGN AND ENGINEERING. REPLACEMENT SHALL BE IN ACCORDANCE WITH THE TABLE BELOW.

CROSS REFERENCE			
CANCELLED			SUPERSEDED BY
MS24650	AN504		MS24630
DASH NO.	DASH NO.	DASH NO.	DASH NO.
1			2
2			3
3			4
10			9
11	C4-4	C4R4	10
12	C4-5	C4R5	11
13	C4-6	C4R6	12
14	C4-8	C4R8	13
15	C4-10	C4R10	14
16	C4-12	C4R12	15
21			20
22	C6-4	C6R4	21
23	C6-5	C6R5	22
24	C6-6	C6R6	23
25	C6-8	C6R8	24
26	C6-10	C6R10	25
27	C6-12	C6R12	26
28	C6-14	C6R14	27
29	C6-16	C6R16	28
32	C8-4	C8R4	33
33	C8-5	C8R5	34
34	C8-6	C8R6	35
35	C8-8	C8R8	36
36	C8-10	C8R10	37
37	C8-12	C8R12	38
38	C8-14	C8R14	39
39	C8-16	C8R16	40
	C8-20	C8R20	41
44	C10-5	C10R5	44
45	C10-6	C10R6	45
46	C10-8	C10R8	46
47	C10-10	C10R10	47
48	C10-12	C10R12	48
49	C10-14	C10R14	49
50	C10-16	C10R16	50
51	C10-20	C10R20	51
52	C10-24	C10R24	52
55			56
56	C416-8	C416R8	57
57	C416-10	C416R10	58
58	C416-12	C416R12	59
59	C416-14	C416R14	60
60	C416-16	C416R16	61
61	C416-20	C416R20	62
62	C416-24	C416R24	63

CROSS REFERENCE			
CANCELLED			SUPERSEDED BY
MS24650	AN504		MS24630
DASH NO.	DASH NO.	DASH NO.	DASH NO.
	C1032-5	C1032R5	64
	C1032-6	C1032R6	65
	C1032-8	C1032R8	66
	C1032-10	C1032R10	67
	C1032-12	C1032R12	68
	C1032-14	C1032R14	69
	C1032-16	C1032R16	70
	C1032-20	C1032R20	71
	C1032-24	C1032R24	72
	C428-8	C428R8	73
	C428-10	C428R10	74
	C428-12	C428R12	75
	C428-14	C428R14	76
	C428-16	C428R16	77
	C428-20	C428R20	78
	C428-24	C428R24	79

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AIR FORCE-82		
PROCUREMENT SPECIFICATION FF-S-107	SUPERSEDES: MS24650 AND AN504	SHEET 2 OF

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